DEVELOPER GUIDE

# What Clinicians Want You to Build

Clinician-defined priorities for AI in primary care

**Event:** AI in Healthcare Co-Design / Hackathon | June 20, 2026 | Invest Ottawa
**Purpose of this guide:** To share what primary care clinicians have told us they need from AI, so your team can build solutions that address real clinical problems.

## Background: Why this guide exists

Primary care in Canada is under severe pressure. Family physicians lose roughly 19 hours per week to administrative work, over half report burnout, and nearly one in five Canadians has no regular primary care provider. AI tools have enormous potential to help, but the history of electronic health records offers a cautionary lesson: technology built without clinician input often increases burden rather than reducing it.

To avoid repeating that mistake, our research team has spent the past year running a structured priority-setting program called **COMPASS** (Consensus On Medical Priorities and AI Solutions in Primary Care). Through facilitated sessions with healthcare AI experts and practicing primary care physicians, we have identified the problems clinicians most want solved and the conditions under which they would actually use AI tools in daily practice.

This guide distils those findings into a practical reference for your team. The challenge areas and design requirements described here are drawn directly from what clinicians told us. On event day, clinician subject matter experts will be available to consult with your team, answer questions about workflows, and help you refine your approach.

> **The core message from clinicians**
> Clinicians are not asking AI to replace clinical judgement. They want tools that make primary care more doable by reducing repetitive work, supporting preparation and follow-through, and protecting the continuity of patient care.

## Challenge areas: What to build

The following domains were identified by clinicians as high-priority areas where AI could deliver meaningful value. Each represents a real workflow problem that consumes clinician time, introduces risk of errors or missed follow-ups, or fragments patient care. These are your starting points.

### 1. Pre-visit preparation

Clinicians described visits that start cold because they lack time to review the chart. They want AI that can scan a patient's record before the appointment and surface what matters: what's overdue, what changed since the last visit, what the patient's agenda might be, and which questionnaires or screening tools should be ready.

- Chart scan that flags items due (e.g., vaccinations, cancer screening, chronic disease monitoring)
- Contextual summary of what has changed since the last visit, including incoming results, hospital reports, and specialist notes
- Dynamic selection of pre-visit questionnaires based on the patient's active problems and visit reason
- Patient-facing pre-visit check-in that captures the agenda and relevant updates before the patient arrives

### 2. Encounter support and point-of-care reasoning

During the visit, clinicians want contextual nudges that surface the right information at the right moment without overwhelming them. This goes well beyond ambient documentation. They described wanting prompts that connect the patient's history to current guidelines, flag potential interactions, and retrieve relevant community resources, all without adding clicks or disrupting flow.

- Guideline-based prompts during the visit (e.g., "this patient has heart failure — have you considered Jardiance?")
- Smart search across scanned documents, faxes, and external reports to retrieve relevant information on the spot
- Community resource retrieval (e.g., local mental health supports, physiotherapy, social services)
- Tailored information summaries that account for the patient's full problem list, not just the presenting issue

### 3. Medication and prescribing support

Medication workflows were among the most prominent pain points. Clinicians described two distinct needs: safety (catching interactions, supporting deprescribing, ensuring monitoring) and logistics (limited-use codes, drug shortages, cost lookups, pharmacy coordination).

- Polypharmacy review and deprescribing suggestions for patients on multiple chronic medications
- Monitoring reminders for high-risk therapies (e.g., blood work required for psychiatric or rheumatologic medications)
- Automated limited-use code reminders so clinicians don't miss coverage eligibility for their patients

- Real-time drug shortage and back-order alerts at the point of prescribing
- Pharmacy cost comparison and coverage status checks before the prescription is sent
- Cross-setting medication reconciliation, especially for patients seen in hospital or by multiple specialists

### 4. Follow-up and care plan automation

Clinicians described post-visit work as a major source of risk and frustration. Tasks that should happen after the encounter, like test tracking, referral follow-up, or patient reminders, frequently fall through the cracks because no system reliably converts visit decisions into downstream actions.

- Automated follow-up task generation from visit notes (e.g., “book follow-up in 6 weeks,” “track CBC result”)
- Test completion tracking with alerts when expected results have not returned
- Patient reminder systems tied to care plan milestones (e.g., medication titration steps, screening intervals)
- Complex care plan scaffolding for patients with multiple active conditions and overlapping follow-up requirements

### 5. Administrative and coordination automation

This was consistently identified as where AI should start. Clinicians described a daily landscape of forms, inbox messages, faxes, billing tasks, and referral paperwork that consumes time without adding clinical value. AI experts separately ranked a “digital medical office assistant” as the single highest-priority project idea.

- Inbox and fax triage: classify, deduplicate, route, and summarize incoming messages so clinicians see what needs attention
- Referral intelligence: specialist availability, wait times, scope-of-practice matching, and automated resubmission for rejected referrals
- Forms automation: auto-populate common paperwork (insurance forms, disability applications, school notes) from existing chart data
- Billing support and roster reconciliation to reduce clerical work for medical office staff
- After-hours triage support for patient calls and messages

### 6. Continuity and whole-person intelligence

Perhaps the most distinctive theme from the clinician session: participants wanted AI that preserves the patient story across time, settings, and providers. Current records were described as fragmented and difficult to navigate. Clinicians imagined tools that reconstruct usable context from years of chart data, including social and family history, and make it available at the point of care.

- Longitudinal patient summaries that capture the narrative arc, not just the problem list
- Family relationship visualization showing connections, shared conditions, and caregiving context
- Social and community context surfacing (e.g., housing, employment, supports) alongside clinical data
- Continuity handoff tools for when patients see a covering physician or cross between care settings

## Design requirements: What makes a tool usable

Strong clinical logic is necessary but not sufficient. Clinicians were clear that how a tool integrates into their day matters as much as what it does. The following requirements emerged repeatedly across both expert and clinician sessions and should guide your design decisions.

| Requirement | What this means for your build |
|---|---|
| **Integration over fragmentation** | Clinicians do not want another standalone app with another login. Design for integration into existing workflows. Fewer clicks, fewer tabs, fewer subscriptions. One clinician warned: “if we end up with 20 different tools and 20 more logins, we’ve failed.” |
| **Low cognitive burden** | Prompts and outputs must be concise and timely. Include a concept of “don’t interrupt me now” — clinicians want control over when and how AI communicates. Relevance filtering matters more than comprehensiveness. |
| **Actionable outputs** | Don’t just generate text. Help convert decisions into tasks: draft the referral, queue the follow-up, prepare the form. The gap between information and action is where care falls through the cracks. |
| **Calibrated trust** | Be transparent about confidence and limitations. Clinicians want to audit, edit, and override. Show your reasoning. Augment judgement, don’t bypass it. |
| **Proactive, not reactive** | The best tools anticipate needs before and after the visit rather than waiting for the clinician to ask. Flag what’s due, track what’s pending, remind what’s unfinished. |
| **Practical affordability** | Small primary care clinics operate on tight margins. Your concept should be realistic about cost. Tools that require expensive infrastructure or per-seat licensing face an immediate adoption barrier. |
| **Privacy by design** | All event materials use synthetic data only. But think ahead: how would your tool handle real patient data? Consider data minimization, local processing, and clear governance from the start. |

## Where experts said to start

In a separate session, nine healthcare AI experts independently ranked near-term project ideas for primary care. Their top picks overlap heavily with the clinician priorities above, reinforcing that these are not speculative ideas but convergent needs. The highest-ranked concepts were:

| # | Project idea | What it does |
|---|---|---|
| 1 | **Digital medical office assistant** | Understands and completes common paperwork, referral packages, scheduling, and medication reviews with structured, auditable outputs. |
| 2 | **Inbox manager** | Ingests EMR messages (labs, hospital reports, pharmacy, insurer), removes duplicates, auto-tags and triages to the right team member, and summarizes threads with proposed actions. |

| # | Project idea | What it does |
| --- | --- | --- |
| 3 | **Ambient scribe + micro-prompts** | Generates structured documentation from the clinical conversation while providing brief, nonintrusive prompts for differentials, guidelines, interaction checks, and one-click actions. |
| 4 | **Pharmacy agent** | Automates coverage checks (limited-use criteria), clarification loops with pharmacies, refill workflows, and drug interaction checks across the prescribing chain. |
| 4 | **Validated triage and self-management** | Patient-facing risk and acuity classification that routes to self-care, other professionals, or primary care with safety nets and clear escalation. |

You do not need to build one of these exact ideas. They are convergence points, not constraints. The challenge domains in the previous section give you the full landscape of clinician needs. Pick the problem that interests your team and that you can prototype credibly in the time available.

## Data and tools for prototyping

All prototyping at this event must use synthetic or de-identified data. No real patient data, no live EMR connections. The following open-access resources can help you build realistic prototypes without any privacy risk. You do not need to use all of these — pick what fits your concept.

### Synthetic patient records

**Synthea — Synthetic Patient Population Simulator**

Synthea is an open-source tool from MITRE that generates realistic (but entirely fictional) patient records, including conditions, encounters, medications, observations, care plans, and procedures. It models disease progression over a patient's lifetime using real-world statistics from the CDC and NIH. Records export in HL7 FHIR (R4), C-CDA, and CSV formats. You can generate a handful of patients or a million. Pre-built datasets of 100 and 1,000 patients are available for immediate download, and bulk datasets of one million records are also available.

- Website: https://synthetichealth.github.io/synthea/
- GitHub: https://github.com/synthetichealth/synthea
- Pre-built downloads (FHIR R4, C-CDA, CSV): https://synthea.mitre.org/downloads
- Best for: any prototype that needs patient timelines, encounter histories, medication lists, lab results, or care plans

### De-identified clinical data

**MIMIC-IV — Medical Information Mart for Intensive Care**

MIMIC-IV is a large, freely accessible de-identified dataset from Beth Israel Deaconess Medical Center covering over a decade of hospital admissions (2008–2019). It includes patient measurements, orders, diagnoses, procedures, treatments, and de-identified clinical notes. While it is hospital- and ICU-focused rather than primary care, it is useful for prototypes involving medication reconciliation, discharge summaries, lab result interpretation, or clinical note processing. Access requires free credentialing through PhysioNet. A 100-patient demo subset is available without credentialing for workshops and rapid prototyping.

- Full dataset (requires credentialing): https://physionet.org/content/mimiciv/
- 100-patient demo (open access): https://physionet.org/content/mimic-iv-demo/
- Best for: clinical notes, medication data, lab values, discharge workflows

**MTSamples — Medical Transcription Samples**

A public-domain collection of over 5,000 sample medical transcription reports across 40 specialties, including general medicine, office notes, SOAP/chart/progress notes, and consult/history-and-physical reports. These are realistic clinical narratives, not structured data, making them useful for prototypes that process, summarize, or extract information from clinical text.

- Website: https://mtsamples.com/
- Kaggle download (CSV): https://www.kaggle.com/datasets/tboyle10/medicaltranscriptions
- Best for: clinical note summarization, NLP on medical text, inbox triage prototypes, documentation support tools

## Canadian health reference data

### Health Canada Drug Product Database (DPD)

The DPD contains product-specific information on all drugs approved for use in Canada, including brand names, drug identification numbers (DINs), active ingredients, routes of administration, pharmaceutical forms, therapeutic classifications, and company information. The data extract is available as CSV files and is updated regularly. Useful for any prototype involving prescribing, medication lookups, or drug information.

- Data extract: https://www.canada.ca/en/health-canada/services/drugs-health-products/drug-products/drug-product-database/what-data-extract-drug-product-database.html
- Open data portal: https://open.canada.ca/data/en/dataset/bf55e42a-63cb-4556-bfd8-44f26e5a36fe
- Best for: medication-related prototypes, drug interaction lookups, prescribing support tools

### Canadian Clinical Drug Data Set (CCDD)

The CCDD is a standardized drug terminology and coding system designed for use in electronic prescribing, EMR systems, and medication reconciliation. It provides standardized drug names, codes, and special groupings (e.g., controlled substances) aligned with international terminology standards. It complements the DPD with a structure designed for interoperability across digital health systems.

- Open data portal: https://open.canada.ca/data/en/dataset/3e0a7b9e-a5e9-4131-bde4-ac685a1f1a38
- Best for: structured medication data for electronic prescribing or medication reconciliation prototypes

## FHIR test servers and sandboxes

If your prototype involves querying or writing structured health data, these open FHIR servers let you test against realistic APIs without any setup or credentialing.

| Server | URL | Notes |
|---|---|---|
| **HAPI FHIR (R4)** | https://hapi.fhir.org/ | Open-source, full CRUD, pre-loaded with test data. Widely used for development and testing. |
| **SMART on FHIR Sandbox** | https://launch.smarthealthit.org/ | Supports SMART App Launch for building EHR-integrated apps. Includes sample patient data. |
| **Synthea on HAPI** | (load Synthea output into HAPI) | Generate patients with Synthea, upload FHIR bundles to HAPI for a populated test server. |

## Interoperability standards worth knowing

You do not need to implement these standards for a hackathon prototype, but awareness of them will strengthen your concept and make it more credible to clinician judges evaluating workflow fit.

- **HL7 FHIR (R4):** The dominant standard for health data exchange. If your prototype reads or writes patient data, FHIR is the format to target. Resources like Patient, Condition, MedicationRequest, Observation, and Encounter map directly to the clinical workflows described in this guide. https://hl7.org/fhir/R4/
- **CDS Hooks:** An open standard for embedding clinical decision support into EHR workflows. If your concept involves triggering prompts or recommendations at specific moments (e.g., when a prescription is signed or a patient chart is opened), CDS Hooks defines how that integration works. https://cds-hooks.hl7.org/
- **SMART on FHIR:** A framework for launching third-party apps from within an EHR. If your concept is an app that a clinician would use inside their medical record system, SMART on FHIR defines the launch, authentication, and data access patterns. https://docs.smarthealthit.org/

**Prototype tip**

For the hackathon, focus on demonstrating the clinical concept clearly. A working prototype with synthetic data and a credible integration story will score higher than a polished front-end with no clinical logic. Clinician judges care about whether the tool solves a real problem and could fit their workflow.

## Practical notes for event day

**Clinician subject matter experts** will be available throughout the day. They can help you understand workflows, validate assumptions, and pressure-test your concept. Use them early and often — their input is the most valuable resource in the room.

**Synthetic data only.** All session materials use fictional, non-identifiable examples. No real patient data, no EMR integration, and no live clinical systems. Design your prototype with synthetic scenarios.

**Judging criteria.** Prototypes will be scored by a clinician panel across four dimensions: (1) clinical importance — does it address a real, high-frequency problem? (2) technical feasibility — could it realistically be built and deployed within 6–12 months? (3) safety and privacy — what is the risk profile and what governance would it need? (4) workflow fit — how likely is it to integrate into existing primary care workflows without adding burden?

**What happens next.** The most promising concepts from this event will be prioritized for follow-on development and potential piloting in real clinical settings through our research partnerships with the Bruyère Health Research Institute and the University of Ottawa Department of Family Medicine. This is not a one-day exercise — it is the entry point to a pipeline that can take strong ideas into clinical testing.

> **To developers**
> Primary care clinicians have told us what they need. They are ready to partner with you to build it. The opportunity is to create tools that are designed with clinicians from the start — not retrofitted after the fact.

### About this guide

This guide is based on findings from the COMPASS program (Consensus On Medical Priorities and AI Solutions in Primary Care), a multi-phase consensus initiative led by the University of Ottawa Department of Family Medicine and The Ottawa Hospital Department of Family Practice. COMPASS uses structured sessions with AI experts, primary care clinicians, and health system leaders to identify which AI capabilities and use cases should be prioritized for development and evaluation in primary care. COMPASS is a component of NAVIGATOR (Needs-driven AI Vision, Innovation, & Governance Accelerator for Transforming Primary Care).

For questions about the event or this guide, contact Dr. Arun Radhakrishnan (aradhakrishnan@toh.ca), Zack van Allen, PhD, MBA (zvanalle@uottawa.ca), or Blake Daly, MBA (bdaly@bruyere.org).